

**Important Information. Please read carefully.**



### **Composition**

Each capsule contains Loperamide Hydrochloride 2mg.

### **Pharmacodynamics**

Loperamide binds to the opiate receptor in the gut wall, reducing propulsive peristalsis, increasing intestinal transit time and enhancing resorption of water and electrolytes. Loperamide increases the tone of the anal sphincter, which helps reduce faecal incontinence and urgency.

### **Pharmacokinetics**

#### *Absorption*

Most ingested loperamide is absorbed from the gut, but as a result of significant first pass metabolism, systemic bioavailability is only approximately 0.3%.

#### *Distribution*

Studies on distribution in rats show a high affinity for the gut wall with a preference for binding to receptors of the longitudinal muscle layer. The plasma protein binding of loperamide is 95%, mainly to albumin. Non-clinical data have shown that loperamide is a P-glycoprotein substrate.

#### *Metabolism*

Loperamide is almost completely extracted by the liver, where it is predominantly metabolized, conjugated and excreted via the bile. Oxidative N-demethylation is the main metabolic pathway for loperamide, and is mediated mainly through CYP3A4 and CYP2C8. Due to this very high first pass effect, plasma concentrations of unchanged drug remain extremely low.

#### *Elimination*

The half-life of loperamide in man is about 11 hours with a range of 9-14 hours. Excretion of the unchanged loperamide and the metabolites mainly occurs through the faeces.

### **Indication**

Loperamide Hydrochloride is indicated for the symptomatic control of acute and chronic diarrhoea. In patients with an ileostomy it can be used to reduce the number and volume of stools and to harden their consistency.

### **Recommended Dosage**

To be administered orally.

#### Adults and Children 6 to 17 Years

The capsules should be taken with liquid

*Acute diarrhoea:* The initial dose is 2 capsules (4 mg) for adults and 1 capsule (2 mg) for children; followed by 1 capsule (2 mg) after every subsequent loose stool.

*Chronic diarrhoea:* The initial dose is 2 capsules (4 mg) daily for adults and 1 capsule (2 mg) daily for children; this initial dose should be adjusted until 1-2 solid stools a day are obtained, which is usually achieved with a maintenance dose of 1-6 capsules (2 mg - 12 mg) daily.

The maximum dose for acute and chronic diarrhoea is 8 capsules (16 mg) daily for adults; in children it must be related to the body weight (3 capsules/20 kg) but should not exceed a maximum of 8 capsules per day.

#### Elderly

No dose adjustment is required for the elderly.

#### Renal Impairment

No dose adjustment is required for patients with renal impairment.

#### Hepatic Impairment

Although no pharmacokinetic data are available in patients with hepatic impairment, Loperamide Hydrochloride should be used with caution in such patients because of reduced first pass metabolism.

### **Contraindications**

This medicine is contraindicated:

- hypersensitivity to the active substance or to any of the excipients.
- in children less than 12 years of age.
- in patients with acute dysentery, which is characterised by blood in stools and high fever.
- in patients with acute ulcerative colitis.
- In patients with bacterial enterocolitis caused by invasive organisms including Salmonella, Shigella and Campylobacter.
- in patients with pseudomembranous colitis associated with the use of broad-spectrum antibiotics.

Loperax must not be used when inhibition of peristalsis is to be avoided due to the possible risk of significant sequelae including ileus, megacolon and toxic megacolon. Loperax must be discontinued promptly when ileus, constipation or abdominal distension develop.

### **Warning and Precautions**

Not recommended for children under 6 years of age. Its use has been associated with fatal episodes of paralytic ileus in infants and young children.

Treatment of diarrhoea with Loperax is only symptomatic. Whenever an underlying etiology can be determined, specific treatment should be given when appropriate. The priority in acute diarrhoea is the prevention or reversal of fluid and electrolyte depletion. This is particularly important in young children and in frail and elderly patients with acute diarrhoea. Use of this medicine does not preclude the administration of appropriate fluid and electrolyte replacement therapy.

Appropriate fluid and electrolyte therapy should be given to protect against dehydration in all cases of diarrhoea. Oral rehydration therapy which is the use of appropriate fluids including oral rehydration salts remains the most effective treatment for dehydration due to diarrhoea. The intake of as much of these fluids as possible is therefore imperative. Drug-induced inhibition of peristalsis may result in fluid retention in the intestine, which may aggravate and mask dehydration and depletion of electrolytes. If severe dehydration or electrolyte imbalance is present, loperamide should be withheld until appropriate corrective therapy has been initiated.

Since persistent diarrhoea can be an indicator of potentially more serious conditions, this medicine should not be used for prolonged periods until the underlying cause of the diarrhoea has been investigated.

In acute diarrhoea, if clinical improvement is not observed within 48 hours, the administration of Loperax should be discontinued and patients should be advised to consult their doctor.

Patients with AIDS treated with this medicine for diarrhoea should have therapy stopped at the earliest signs of abdominal distension. There have been isolated reports of obstipation with an increased risk for toxic megacolon in AIDS patients with infectious colitis from both viral and bacterial pathogens treated with loperamide hydrochloride.

Although no pharmacokinetic data are available in patients with hepatic impairment, this medicine should be used with caution in such patients because of reduced first pass metabolism, as it may result in a relative overdose leading to CNS toxicity.

Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine because it contains lactose.

If patients are taking this medicine to control episodes of diarrhoea associated with Irritable Bowel Syndrome previously diagnosed by their doctor, and clinical improvement is not observed within 48 hours, the administration of loperamide hydrochloride should be discontinued and they should consult with their doctor. Patients should also return to their doctor if the pattern of their symptoms changes or if the repeated episodes of diarrhoea continue for more than two weeks.

Cardiac events including QT prolongation and torsades de pointes have been reported in association with overdose. The use of higher than the recommended doses for control of the diarrhea may cause abnormal heart rhythms and serious cardiac events leading to death. However, in adult patients receiving the recommended dosage of loperamide, cases of syncope and ventricular tachycardia have been reported. Some of these patients were taking other drugs or had other risk factors that may have increased their risk of cardiac adverse reactions. Patients should not exceed the recommended dose and/or the recommended duration of treatment.

Only take this medicine to treat acute episodes of diarrhoea associated with Irritable Bowel Syndrome if your doctor has previously diagnosed Irritable Bowel Syndrome.

If any of the following now apply, do not use the product without first consulting your doctor, even if you know you have Irritable Bowel Syndrome:

- If you are aged 40 or over and it is some time since your last Irritable Bowel Syndrome attack
- If you are aged 40 or over and your Irritable Bowel Syndrome symptoms are different this time
- If you have recently passed blood from the bowel
- If you suffer from severe constipation
- If you are feeling sick or vomiting
- If you have lost your appetite or lost weight
- If you have difficulty or pain passing urine
- If you have a fever
- If you have recently travelled abroad

Consult your doctor if you develop new symptoms, or if your symptoms worsen, or your symptoms have not improved over two weeks.

### **Interactions with Other Medicines**

Non-clinical data have shown that loperamide is a P-glycoprotein substrate. Concomitant administration of loperamide (16 mg single dose) with quinidine, or ritonavir, which are both P-glycoprotein inhibitors, resulted an increase in loperamide plasma levels.

The concomitant administration of loperamide (4 mg single dose) and itraconazole, an inhibitor of CYP3A4 and P-glycoprotein, resulted an increase in loperamide plasma concentrations. An inhibitor of CYP2C8, gemfibrozil, increased loperamide. The combination of itraconazole and gemfibrozil resulted an increase in peak plasma levels of loperamide and total plasma exposure.

The concomitant administration of loperamide (16 mg single dose) and ketoconazole, an inhibitor of CYP3A4 and P-glycoprotein, resulted an increase in loperamide plasma concentrations. Concomitant treatment with oral desmopressin resulted an increase of desmopressin plasma concentrations, presumably due to slower gastrointestinal motility.

It is expected that drugs with similar pharmacological properties may potentiate loperamide's effect and that drugs that accelerate gastrointestinal transit may decrease its effect.

### **Pregnancy & Lactation**

#### Pregnancy

Safety in human pregnancy has not been established, although from animal studies there are no indications that loperamide hydrochloride possesses any teratogenic or embryotoxic properties. As with other drugs, it is not advisable to administer this medicine in pregnancy, especially during the first trimester.

### Breast-feeding

Small amounts of loperamide may appear in human breast milk. Therefore, this medicine is not recommended during breast-feeding.

Women who are pregnant or breast feeding infants should therefore be advised to consult their doctor for appropriate treatment.

### Effects on ability to drive and use machines

Loss of consciousness, depressed level of consciousness, tiredness, dizziness, or drowsiness may occur when diarrhoea is treated with this medicine. Therefore, it is advisable to use caution when driving a car or operating machinery.

### Side Effects

Table 1: Adverse Drug Reactions

| System Organ Class                                   | Indication                           |                                                                                                      |                                                                                                                                                  |
|------------------------------------------------------|--------------------------------------|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                      | Common                               | Uncommon                                                                                             | Rare                                                                                                                                             |
| Immune System Disorders                              |                                      |                                                                                                      | Hypersensitivity reaction<br>Anaphylactic reaction (including Anaphylactic shock)<br>Anaphylactoid reaction                                      |
| Nervous System Disorders                             | Headache                             | Dizziness<br>Somnolence                                                                              | Loss of consciousness<br>Stupor<br>Depressed level of consciousness<br>Hypertonia<br>Coordination abnormality                                    |
| Eye Disorders                                        |                                      |                                                                                                      | Miosis                                                                                                                                           |
| Gastrointestinal Disorders                           | Constipation<br>Nausea<br>Flatulence | Abdominal pain<br>Abdominal discomfort<br>Dry mouth<br>Abdominal pain upper<br>Vomiting<br>Dyspepsia | Ileus (including paralytic ileus)<br>Megacolon (including toxic megacolon)<br>Abdominal distension                                               |
| Skin and Subcutaneous Tissue Disorders               |                                      | Rash                                                                                                 | Bullous eruption (including Stevens-Johnson syndrome, Toxic epidermal necrolysis and Erythema multiforme)<br>Angioedema<br>Urticaria<br>Pruritus |
| Renal and Urinary Disorders                          |                                      |                                                                                                      | Urinary retention                                                                                                                                |
| General Disorders and Administration Site Conditions |                                      |                                                                                                      | Fatigue                                                                                                                                          |

Post-marketing experience reported cardiac disorders such as QT/QTc interval prolongation, Torsades de Pointes, other ventricular arrhythmias, cardiac arrest, syncope, and death (see *Warning and Precautions*).

### Overdose & treatment

In case of overdose (including relative overdose due to hepatic dysfunction), CNS depression (stupor, coordination abnormality, somnolence, miosis, muscular hypertonia and respiratory depression), constipation, urinary retention and ileus may occur. Children, and patients with hepatic dysfunction, may be more sensitive to CNS effects.

In individuals who have ingested overdoses of loperamide hydrochloride, cardiac events such as QT interval prolongation, torsades de pointes, other serious ventricular arrhythmias, cardiac arrest and syncope have been observed. Fatal cases have also been reported.

In cases of overdose, ECG monitoring for QT interval prolongation should be initiated.

If CNS symptoms of overdose occur, naloxone can be given as an antidote. Since the duration of action of loperamide is longer than that of naloxone (1 to 3 hours), repeated treatment with naloxone might be indicated. Therefore, the patient should be monitored closely for at least 48 hours in order to detect possible CNS depression.

### **Presentation**

Blue/White hard gelatin capsules with "XSP" printed, in packs of 10's and 100's.

### **Shelf-life**

The expiry date is indicated on the packaging.

### **Storage**

Store in a cool dry place below 30°C.

**KEEP ALL MEDICINES OUT OF REACH OF CHILDREN / JAUHI DARI KANAK-KANAK**

For further information, please consult your pharmacist or physician.

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